

Schematic

(8kW EV WPT Tx Board, Ver 1.0E00)

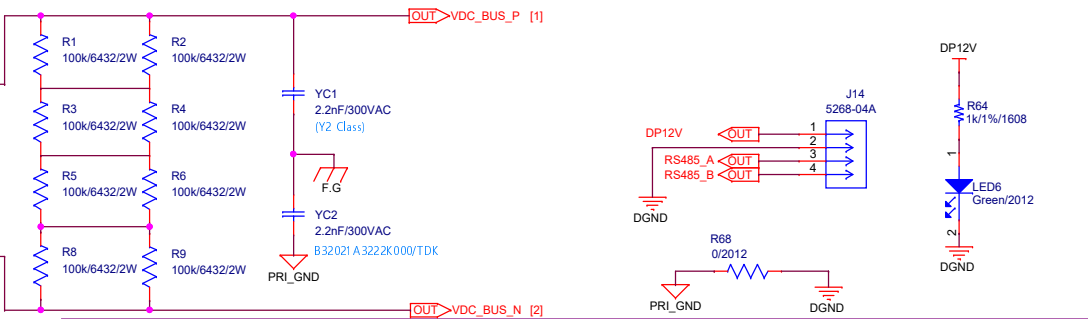
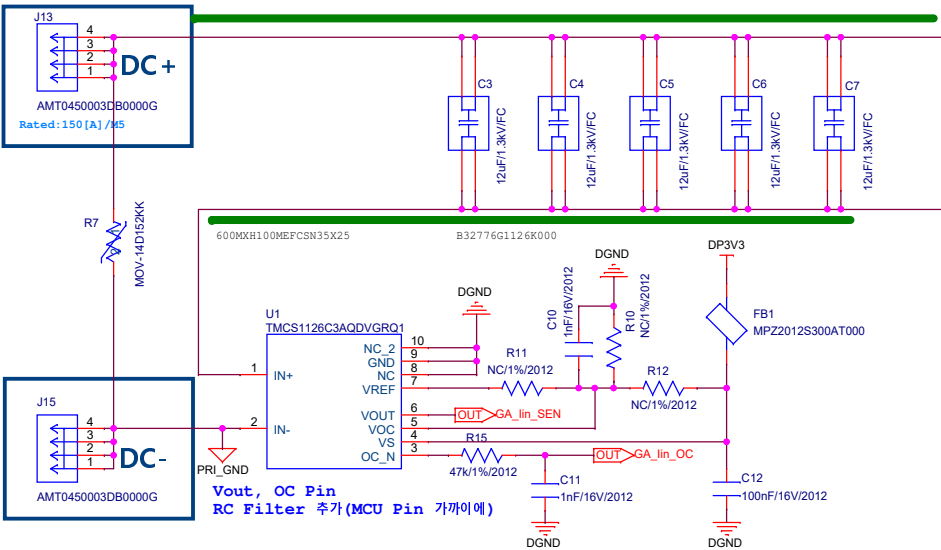
Date : 2026.04.27
Designer : Jong-Hyeon, Oh

Eta Electronics Inc.

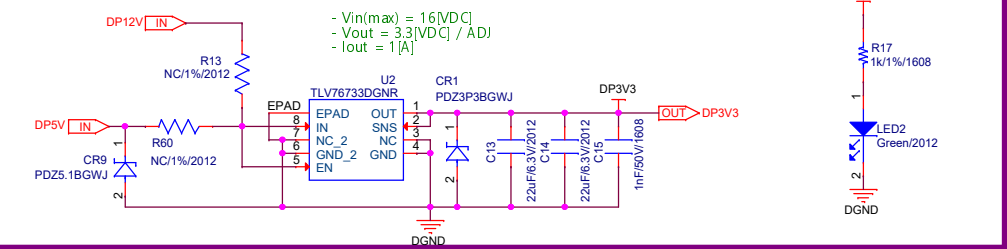
Reference	Description	DSN.BY	CHK.BY	CHK.BY	APPR.BY	MODEL	8kW EV WPT Tx Board	
		J.H.OH	J.H.OH	D.W.KIM	T.D.YEO	Title 8kW EV WPT Tx Board		
						Size A3	Document Number <Doc>	Rev 1.0E00
						2026.04.27	2026.04.27	2026.04.27

입력전원 (200VDC ~ 800VDC)

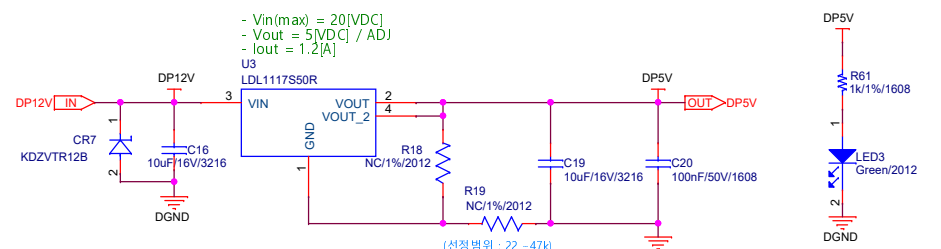
(연결@절연형 DC Power Supply)



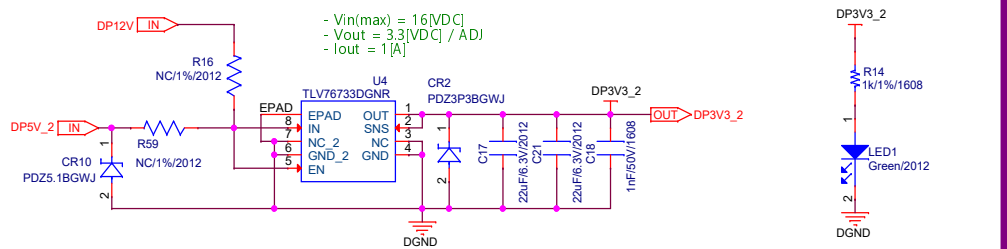
제어전원 (LD0 Voltage Regulator : DP3V3, 1A)



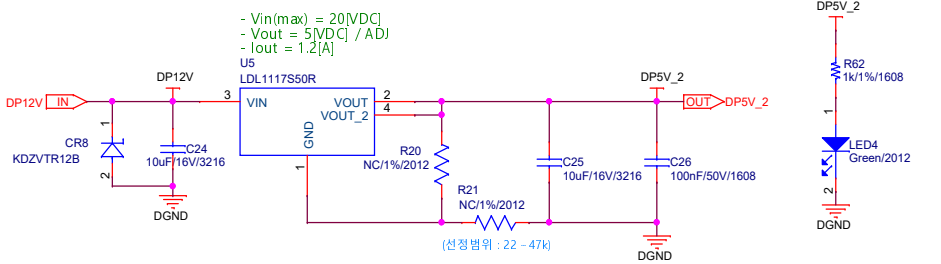
제어전원 (LD0 Voltage Regulator : DP5V, 1.2A)



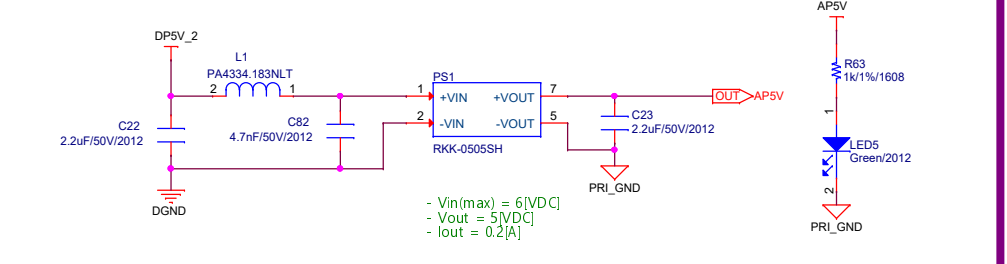
제어전원 (LD0 Voltage Regulator : DP3V3, 1A)



제어전원 (LD0 Voltage Regulator : DP5V, 1.2A)



제어전원 (DC/DC Converter : AP5V, 0.2A)



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1		J.H.OH	J.H.OH	D.W.KIM	T.D.YEO	Title8kW EV WPT Tx Board SizeA3<Doc>Rev1.0E00 Date:Monday, May 04, 2026Sheet2 of 6	
2							
3		2026.04.27	2026.04.27	2026.04.27	2026.04.27		

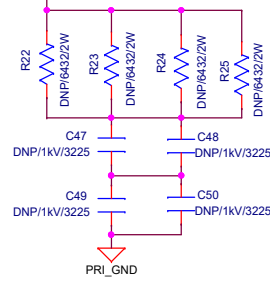
Inverter Power Stack (H- Bridge)

전압 : 200 ~ 800[VDC]
PCB Pattern : ??[Apeak] @??[VDC]

[2] VDC_BUS_P [IN]

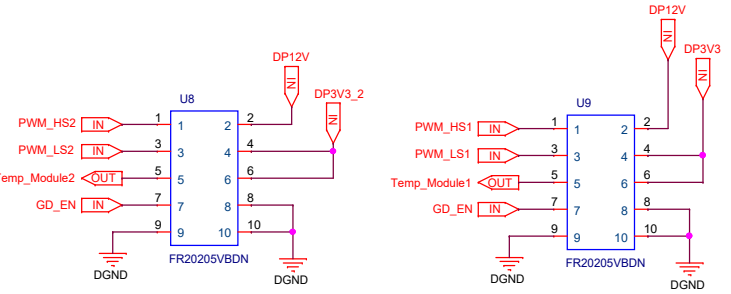
VDC_BUS_P [2]

MOSFET Module 바로 옆



[2] VDC_BUS_N [IN]

PRI_GND



TX_COIL_A

AMT0450003DB0000G

FF8MR12W1M1H_B70

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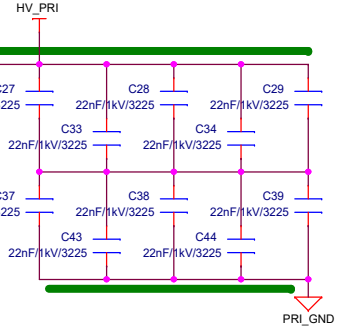
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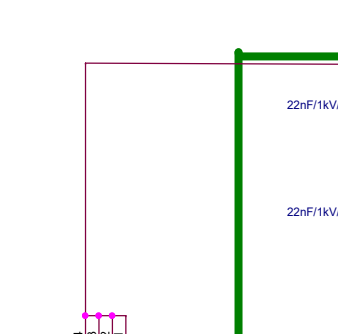
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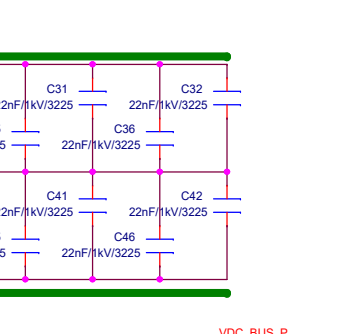
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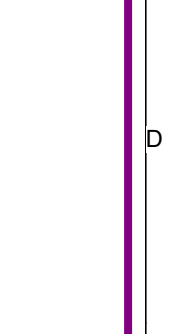
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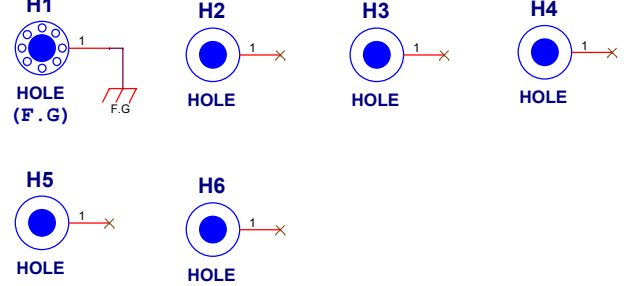
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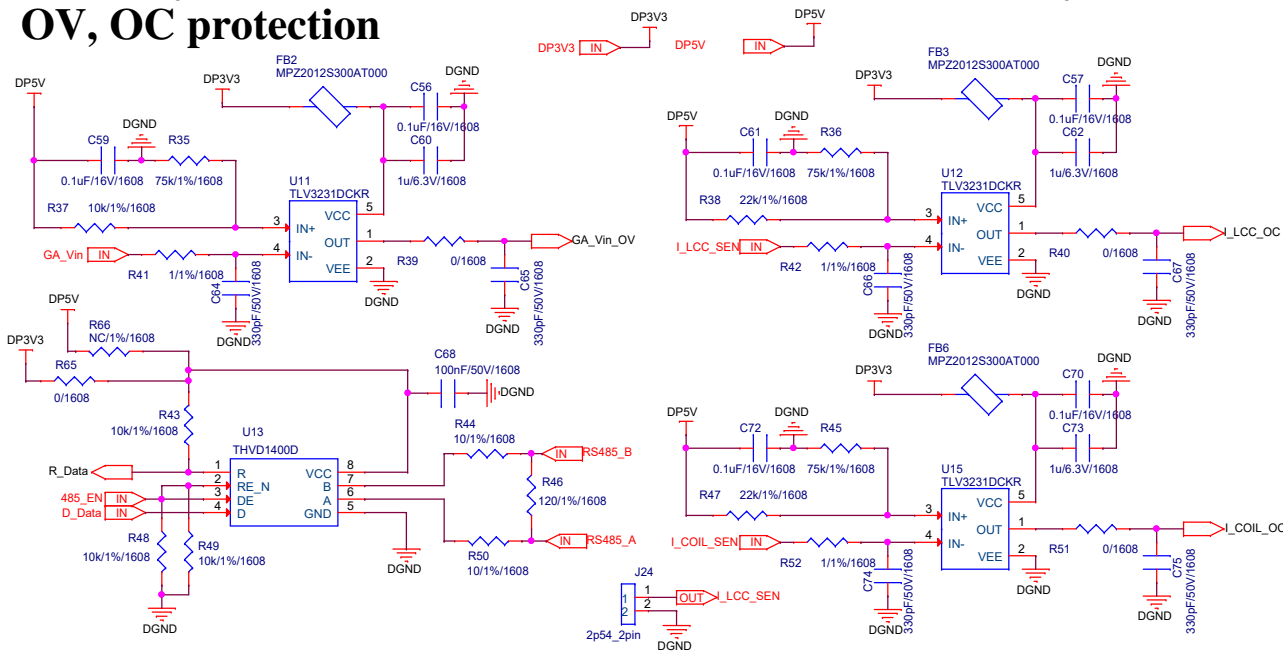
PCB Mounting Hole (M3적용)



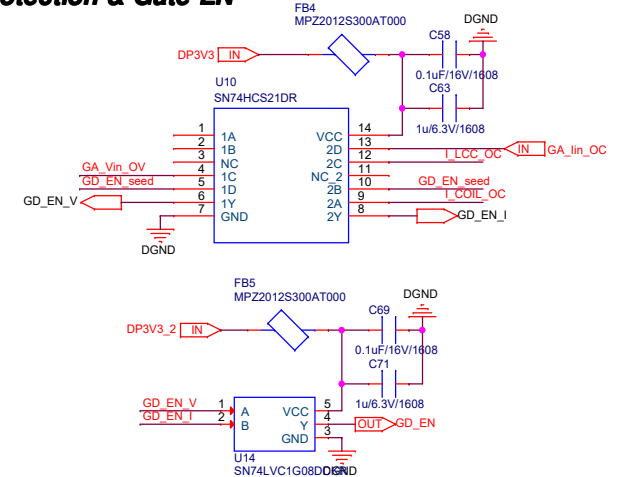
Reference	Description	DSN.BY	CHK.BY	CHK.BY	APPR.BY	MODEL	8kW EV WPT Tx Board
1		J.H.OH	J.H.OH	D.W.KIM	T.D.YEO		
2							
3							

Title		8kW EV WPT Tx Board	
Size	Document Number	Rev 1.0E00	
A3	<Doc>		
Date:	Monday, May 04, 2026	Sheet	3 of 6

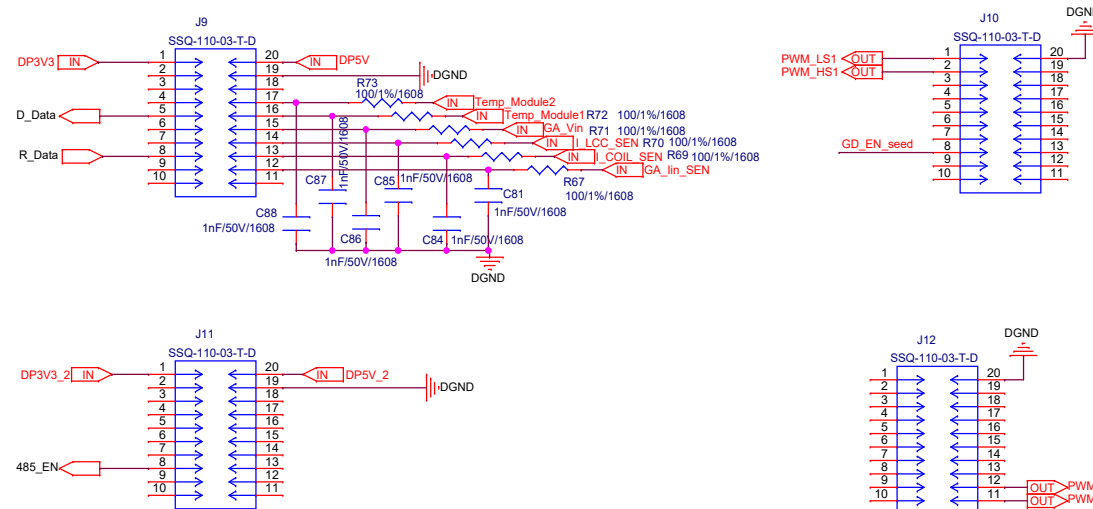
OV, OC protection



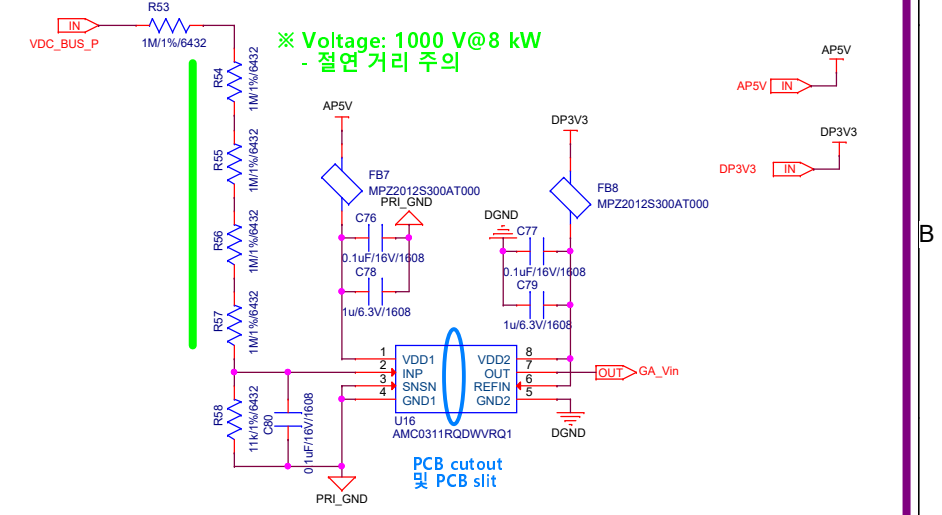
Protection & Gate EN



MCU Pin Connect



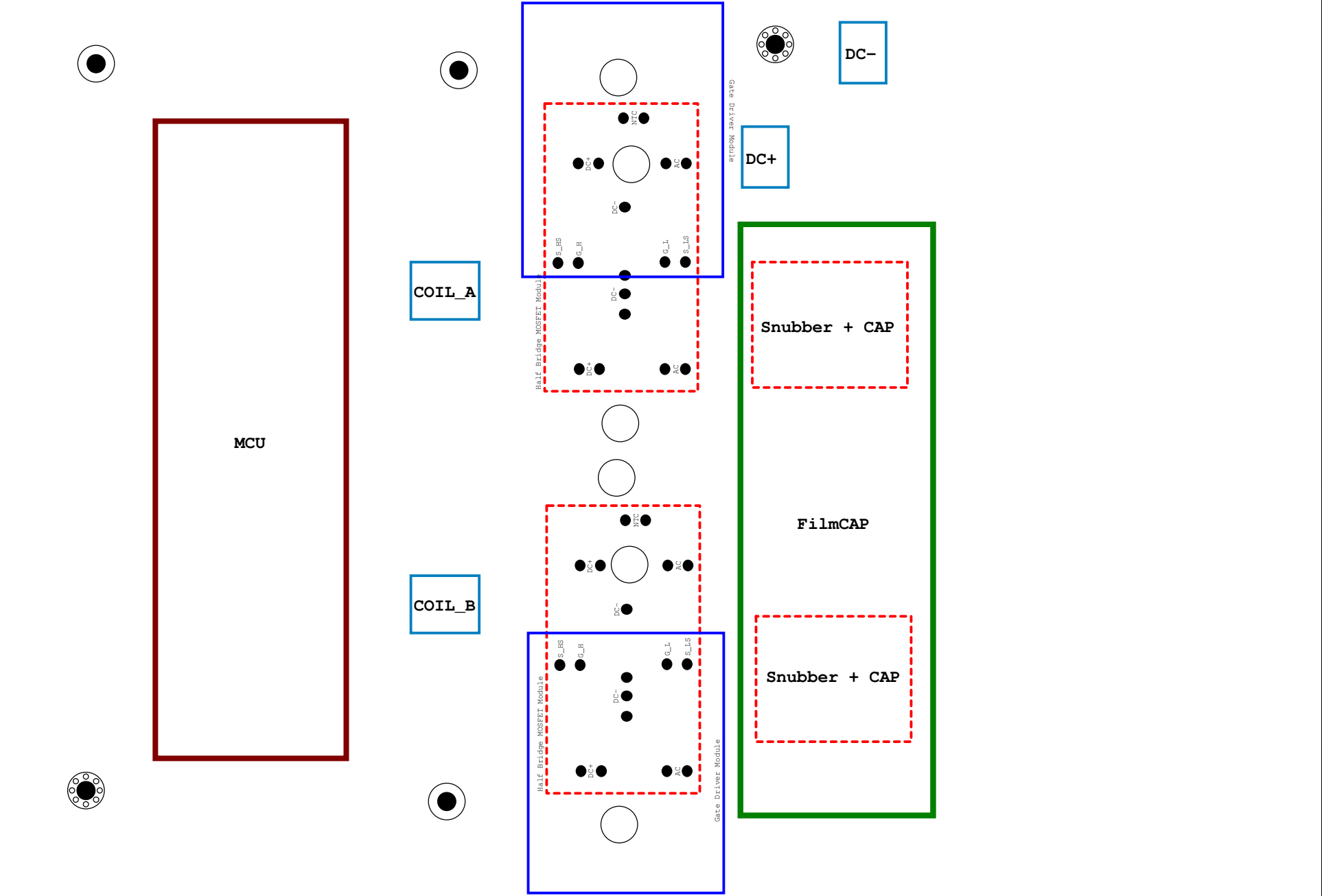
Input Sensing ※ 모든 MLCC 소자 IC 최단 거리 배치



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Title		8kW EV WPT Tx Board	
Size	Document Number	Rev 1.0E00	
A3	<Doc>		
Date:	Thursday, April 30, 2026	Sheet	5 of 6

EV WPT Tx Board 부품배치도



Reference	Description	DSN.BY	CHK.BY	CHK.BY	APPR.BY	MODEL	8kW EV WPT Tx Board	
		J.H.OH	J.H.OH	D.W.KIM	T.D.YEO	Title 8kW EV WPT Tx Board		
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8kW EV WPT Tx Board의 Revision 내용

설계 참조자료

TEST PIN 처리 설명

모양	내용 설명
	Via Hole 처리
	Test Pointer 처리

캐패시터 온도 및 오차

CODE	오차	온도 특성
COG	± 30[ppm]	- 55~ 125[°C]
X7R	± 15[%]	- 55~ 125[°C]
Y5V	± 22~ - 82[%]	- 30~ 85[°C]

칩 저항 SIZE

Type	Power	최대 사용전압
0603 (0201)	0.05[W]	25[V]
1005 (0402)	0.1[W]	50[V]
1608 (0603)	0.1[W]	50[V]
2012 (0805)	0.125[W]	150[V]
3216 (1206)	0.25[W]	200[V]
3225 (1210)	0.5[W]	200[V]
5025 (2010)	0.75[W]	200[V]
6432 (2512)	1.0[W]	200[V]

칩 저항 오차

Type	오차
B	± 0.1[%]
C	± 0.25[%]
D	± 0.5[%]
F	± 1.0[%]
G	± 2.0[%]
J	± 5.0[%]
K	± 10.0[%]
M	± 20.0[%]

 : PCB ROUTER 0.5mm...0.8mm

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						8kW EV WPT Tx Board		
						Size	Document Number	Rev
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		Date: Monday, April 27, 2026				Sheet	7	of 6